

Experiment 32: Implement Pattern Matching

Aim

To write and execute a **Prolog program to implement pattern matching**.

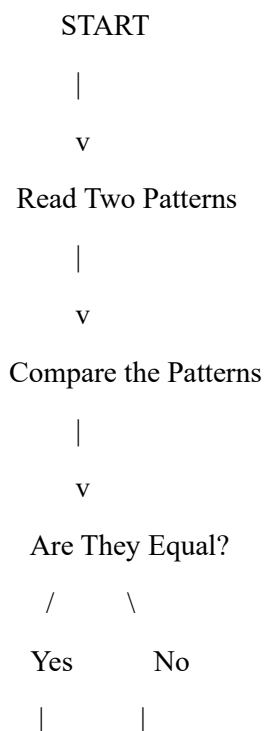
Objectives

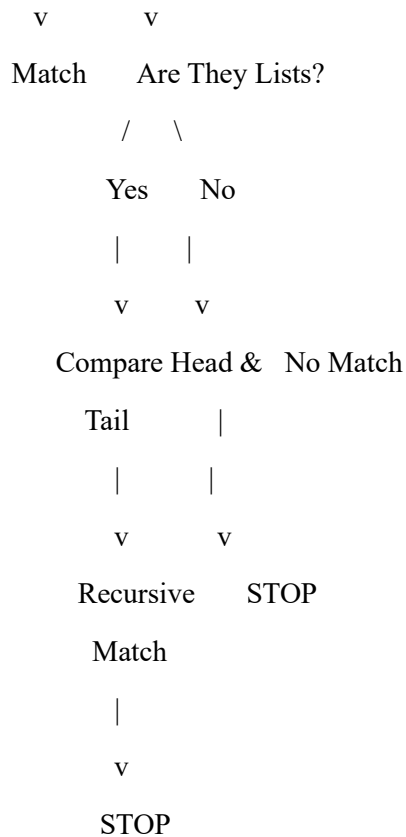
1. To understand pattern matching in Prolog.
2. To learn how Prolog compares structures and lists.
3. To understand unification between terms.
4. To implement pattern matching using Prolog predicates.
5. To verify whether two given patterns match.

Algorithm

1. Start the program.
2. Define a predicate `pattern_match/2`.
3. Compare the two given terms.
4. If both terms are identical, return true.
5. If the terms are lists, compare their heads and tails recursively.
6. Continue until all elements are matched.
7. If any element does not match, return false.
8. Stop.

Flowchart





Program

% Pattern Matching Program

```
pattern_match(X, X).
```

```
pattern_match([], []).
```

```
pattern_match([H1|T1], [H2|T2]) :-
```

```
    pattern_match(H1, H2),
```

```
    pattern_match(T1, T2).
```

% Test Cases

test :-

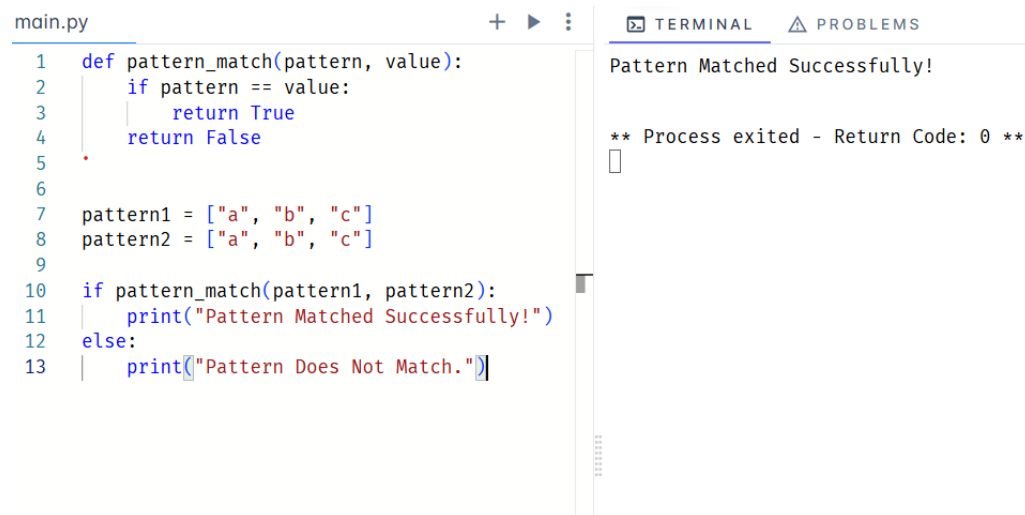
```
    pattern_match([a,b,c], [a,b,c]),
```

```
    write('Pattern Matched Successfully'), nl.
```

test :-

```
\+ pattern_match([a,b,c], [a,b,d]),  
write('Pattern Does Not Match'), nl.
```

Output:



The screenshot shows a code editor with a file named 'main.py'. The code defines a function 'pattern_match' that checks if a pattern list equals a value list. It then defines two lists, 'pattern1' and 'pattern2', both containing ['a', 'b', 'c']. The code calls 'pattern_match(pattern1, pattern2)' and prints 'Pattern Matched Successfully!' if it returns True, or 'Pattern Does Not Match.' if it returns False. To the right of the code editor is a terminal window showing the output 'Pattern Matched Successfully!' and a message '** Process exited - Return Code: 0 **'.

```
main.py  
1 def pattern_match(pattern, value):  
2     if pattern == value:  
3         return True  
4     return False  
5  
6  
7 pattern1 = ["a", "b", "c"]  
8 pattern2 = ["a", "b", "c"]  
9  
10 if pattern_match(pattern1, pattern2):  
11     print("Pattern Matched Successfully!")  
12 else:  
13     print("Pattern Does Not Match.")
```

TERMINAL PROBLEMS
Pattern Matched Successfully!

** Process exited - Return Code: 0 **

Result

The Prolog program was successfully executed to implement **pattern matching**.

Conclusion

pattern matching was successfully implemented using Prolog. The experiment demonstrates the use of **unification, recursion, and list processing** in Prolog.